

Simulation Environment

Network Simulator

OMNeT++(OpenCV2X)

Veins
(TraCI)

Traffic Simulator

SUMO

PQC Integration

ECDSA (baseline) and Falcon-512 . Full SPDU size: 305 Byte vs. 1,739 Byte

C-V2X Mode 4 Configuration

20 MHz · 100 RBs · 10 RBs/subchannel · MCS 5–11 · 23 dBm TX · 100 ms RRI, $p_{keep} = 0.8$ · Full certificate in every 5th BSM, digest otherwise following SAE J3161 and IEEE 1609.2

Urban Traffic Scenario Configuration

4-way intersection · Line-of-sight · traffic level-of-service C

Simulation Data Post-processing

PDR Analysis

End-to-end Latency Analysis